

GraphTheorySymbol:

A set of $\text{\LaTeX}$ commands to have symbols designed for
Graph Theory

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Abstract

This documentation is for $\text{\LaTeX}$ package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols were created by the author of this package.

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1 Introduction

This documentation is for $\text{\LaTeX}$ package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols were created by the author of this package.

This package is licenced under Lesser GNU General Public License version 3 or later (LGPLv3+). It means you are free do anything with it, but if you do improve the code of the symbols that we provide, please do share these improvements in a similar way. There is no restriction on other graph theory symbols/code, even if it would be a good fit for inclusion in this package. There is also no need to share in a similar way what you build with this package. The text of the licenses is available at:

<https://www.gnu.org/licenses/>.

The git repository of this source code is also available at:

<https://github.com/LLyaudet/GraphTheorySymbol/>.

Clone it with this URL:

[git@github.com:LLyaudet/GraphTheorySymbol.git](https://github.com:LLyaudet/GraphTheorySymbol.git).

This package is available on CTAN:

<https://ctan.org/pkg/graph-theory-symbol>.

This is our first $\text{\LaTeX}$ package. It may be that our macros are quite slow and that a document making extensive use of these macros will be slow to compile. If you know how to produce better $\text{\LaTeX}$ code, please submit a pull request on GitHub.

2 Usage

2.1 Memo or for the impatient

Add in your preamble: `\usepackage{graph-theory-symbol}`.

Long name	Short name	Result	In context
<code>\GTSadjacency/</code>	<code>\GTSa/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyRelation/</code>	<code>\GTSaR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacency/</code>	<code>\GTSu/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyRelation/</code>	<code>\GTSuR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUp/</code>	<code>\GTSdAU/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpRelation/</code>	<code>\GTSdAUR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUp/</code>	<code>\GTSnDAU/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpRelation/</code>	<code>\GTSnDAUR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDown/</code>	<code>\GTSdAD/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownRelation/</code>	<code>\GTSdADR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDown/</code>	<code>\GTSnDAD/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownRelation/</code>	<code>\GTSnDADR/</code>	$\circ$	$u \circ v$
<code>\GTSincidence/</code>	<code>\GTSi/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceRelation/</code>	<code>\GTSiR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidence/</code>	<code>\GTSnI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceRelation/</code>	<code>\GTSnIR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidence/</code>	<code>\GTSoI/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceRelation/</code>	<code>\GTSoIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidence/</code>	<code>\GTSnOI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceRelation/</code>	<code>\GTSnOIR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidence/</code>	<code>\GTSiI/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceRelation/</code>	<code>\GTSiIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidence/</code>	<code>\GTSnII/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceRelation/</code>	<code>\GTSnIIR/</code>	$\circ$	$v \circ e$
Long name	Short name	Result	In context
<code>\GTSadjacencyMonochrome/</code>	<code>\GTSaM/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyMonochromeRelation/</code>	<code>\GTSaMR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochrome/</code>	<code>\GTSuM/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochromeRelation/</code>	<code>\GTSuMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochrome/</code>	<code>\GTSdAUM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSdAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochrome/</code>	<code>\GTSnDAUM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSnDAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochrome/</code>	<code>\GTSdADM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSdADMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochrome/</code>	<code>\GTSnDADM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSnDADMR/</code>	$\circ$	$u \circ v$
<code>\GTSincidenceMonochrome/</code>	<code>\GTSiM/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceMonochromeRelation/</code>	<code>\GTSiMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochrome/</code>	<code>\GTSnIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochromeRelation/</code>	<code>\GTSnIMR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochrome/</code>	<code>\GTSoIM/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochromeRelation/</code>	<code>\GTSoIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochrome/</code>	<code>\GTSnOIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochromeRelation/</code>	<code>\GTSnOIMR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochrome/</code>	<code>\GTSiIM/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochromeRelation/</code>	<code>\GTSiIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochrome/</code>	<code>\GTSnIIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochromeRelation/</code>	<code>\GTSnIIMR/</code>	$\circ$	$v \circ e$

\GTSsetMainColor{blue}			
Long name	Short name	Result	In context
\GTSadjacency/	\GTSa/		u v
\GTSadjacencyRelation/	\GTSaR/		u v
\GTSunadjacency/	\GTSu/		u v
\GTSunadjacencyRelation/	\GTSuR/		u v
\GTSdirectedAdjacencyUp/	\GTSdAU/		u v
\GTSdirectedAdjacencyUpRelation/	\GTSdAUR/		u v
\GTSnegatedDirectedAdjacencyUp/	\GTSnDAU/		u v
\GTSnegatedDirectedAdjacencyUpRelation/	\GTSnDAUR/		u v
\GTSdirectedAdjacencyDown/	\GTSdAD/		u v
\GTSdirectedAdjacencyDownRelation/	\GTSdADR/		u v
\GTSnegatedDirectedAdjacencyDown/	\GTSnDAD/		u v
\GTSnegatedDirectedAdjacencyDownRelation/	\GTSnDADR/		u v
\GTSincidence/	\GTSi/		v e
\GTSincidenceRelation/	\GTSiR/		v e
\GTSnegatedIncidence/	\GTSnI/		v e
\GTSnegatedIncidenceRelation/	\GTSnIR/		v e
\GTSoutIncidence/	\GTSOI/		v e
\GTSoutIncidenceRelation/	\GTSOIR/		v e
\GTSnegatedOutIncidence/	\GTSnOI/		v e
\GTSnegatedOutIncidenceRelation/	\GTSnOIR/		v e
\GTSinIncidence/	\GTSiI/		v e
\GTSinIncidenceRelation/	\GTSiIR/		v e
\GTSnegatedInIncidence/	\GTSnII/		v e
\GTSnegatedInIncidenceRelation/	\GTSnIIR/		v e

2.2 Detailed usage

First you need to add in your preamble: `\usepackage{graph-theory-symbol}`.

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{\langle commandname \rangle}` The name of the command to create.

`{\langle commandcode \rangle}` The code of the command to create.

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

`{\langle color \rangle}` The color that must be used by GraphTheorySymbol afterward.

`\GTSsetSecondColor` This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

`{\langle color \rangle}` The color that must be used by GraphTheorySymbol afterward.

`\GTS@a` This macro will draw any of the adjacency symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and `\GTS@b`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

`{\langle mainColor \rangle}` If non-empty, this color will be used as main color.

`{\langle secondColor \rangle}` If non-empty, this color will be used as second color.

`{\langle drawSlash \rangle}` Non-empty implies that a negation will be drawn.

`{\langle drawUpArrow \rangle}` Non-empty implies that an up-arrow will be drawn.

`{\langle drawDownArrow \rangle}` Non-empty implies that a down-arrow will be drawn.

`\GTS@b` This macro builds on top of `\GTS@a` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@a`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

`{\langle mainColor \rangle}` If non-empty, this color will be used as main color.

`{\langle secondColor \rangle}` If non-empty, this color will be used as second color.

`{\langle drawSlash \rangle}` Non-empty implies that a negation will be drawn.

`{\langle drawUpArrow \rangle}` Non-empty implies that an up-arrow will be drawn.

`{\langle drawDownArrow \rangle}` Non-empty implies that a down-arrow will be drawn.

`{\langle relation \rangle}` Non-empty implies that relation spacing will be added.

`\GTSxa` Versatile macros for adjacency that can be given options.

`\GTSxadj` Argument:

`\GTSxadjacency` [`\langle optionsList \rangle`] Sets the following options if not empty:

mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
up or u,
down or d,
relation or rel or r.

`\GTSadjacency` This macro will draw an adjacency symbol. A first drawing didn’t include the small dots in the middle of the two circles corresponding to the graph vertices. But it was too similar to “spoon” and multimap symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property to remind of the classical textual notation of adjacency relation as Adj(.,.).

A.`\GTSadjacency/A`.

or A.`\GTSadj/A`.

or $A.\text{\GTSa}/A$.
 will yield $A.\text{\GTSa}A$.
 u $\text{\GTSadjacency}/ v$
 or u $\text{\GTSadj}/ v$
 or u $\text{\GTSa}/ v$
 will yield $u \text{\GTSa} v$.
 $\$u \text{\GTSadjacency}/ v\$$
 or $\$u \text{\GTSadj}/ v\$$
 or $\$u \text{\GTSa}/ v\$$
 will yield $u \text{\GTSa} v$.

$\text{\GTSadjacencyRelation}$ This macro will draw an adjacency symbol.
 $\text{\GTSadjRel}$ $\$u\text{\GTSadjacencyRelation}/v\$$
 $\text{\GTSaR}$ or $\$u\text{\GTSadjRel}/v\$$
 or $\$u\text{\GTSaR}/v\$$
 will yield $u \text{\GTSa} v$.

$\text{\GTSunadjacency}$ This macro will draw an unadjacency symbol.
 $\text{\GTSunadj}$ $A.\text{\GTSunadjacency}/A$.
 $\text{\GTSu}$ or $A.\text{\GTSunadj}/A$.
 or $A.\text{\GTSu}/A$.
 will yield $A.\text{\GTSu}A$.
 u $\text{\GTSunadjacency}/ v$
 or u $\text{\GTSunadj}/ v$
 or u $\text{\GTSu}/ v$
 will yield $u \text{\GTSu} v$.
 $\$u \text{\GTSunadjacency}/ v\$$
 or $\$u \text{\GTSunadj}/ v\$$
 or $\$u \text{\GTSu}/ v\$$
 will yield $u \text{\GTSu} v$.

$\text{\GTSunadjacencyRelation}$ This macro will draw an unadjacency symbol.
 $\text{\GTSunadjRel}$ $\$u\text{\GTSunadjacencyRelation}/v\$$
 $\text{\GTSuR}$ or $\$u\text{\GTSunadjRel}/v\$$
 or $\$u\text{\GTSuR}/v\$$
 will yield $u \text{\GTSu} v$.

$\text{\GTSdirectedAdjacencyUp}$ This macro will draw a directed adjacency upward symbol. The operand on the left
 $\text{\GTSdirAdjUp}$ of the symbol should be considered the source of the arc; the operand on the right of the
 $\text{\GTSdAU}$ symbol should be considered the target of the arc.
 $A.\text{\GTSdirectedAdjacencyUp}/A$.
 or $A.\text{\GTSdirAdjUp}/A$.
 or $A.\text{\GTSdAU}/A$.
 will yield $A.\text{\GTSdAU}A$.
 u $\text{\GTSdirectedAdjacencyUp}/ v$
 or u $\text{\GTSdirAdjUp}/ v$
 or u $\text{\GTSdAU}/ v$
 will yield $u \text{\GTSdAU} v$.
 $\$u \text{\GTSdirectedAdjacencyUp}/ v\$$
 or $\$u \text{\GTSdirAdjUp}/ v\$$
 or $\$u \text{\GTSdAU}/ v\$$
 will yield $u \text{\GTSdAU} v$.

$\text{\GTSdirectedAdjacencyUp-Relation}$ This macro will draw a directed adjacency upward symbol with relation spacing in
 math mode.
 $\text{\GTSdirAdjUpRel}$ $\$u\text{\GTSdirectedAdjacencyUpRelation}/v\$$
 $\text{\GTSdAUR}$ or $\$u\text{\GTSdirAdjUpRel}/v\$$
 or $\$u\text{\GTSdAUR}/v\$$
 will yield $u \text{\GTSdAUR} v$.

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency upward symbol.
`AdjacencyUp` $A.\GTSnegatedDirectedAdjacencyUp/A.$
`\GTSnegDirAdjUp` or $A.\GTSnegDirAdjUp/A.$
`\GTSnDAU` or $A.\GTSnDAU/A.$
will yield $A.\frac{\neg}{\rightarrow}A.$
`u \GTSnegatedDirectedAdjacencyUp/ v`
or `u \GTSnegDirAdjUp/ v`
or `u \GTSnDAU/ v`
will yield $u \frac{\neg}{\rightarrow} v.$
 $\$u \GTSnegatedDirectedAdjacencyUp/ v\$$
or $\$u \GTSnegDirAdjUp/ v\$$
or $\$u \GTSnDAU/ v\$$
will yield $u \frac{\neg}{\rightarrow} v.$

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency upward symbol with relation
`AdjacencyUpRelation` spacing in math mode.
`\GTSnegDirAdjUpRel` $\$u\GTSnegatedDirectedAdjacencyUpRelation/v\$$
`\GTSnDAUR` or $\$u\GTSnegDirAdjUpRel/v\$$
or $\$u\GTSnDAUR/v\$$
will yield $u \frac{\neg}{\rightarrow} v.$

`\GTSdirectedAdjacencyDown` This macro will draw a directed adjacency downward symbol. The operand on the left
`\GTSdirAdjDown` of the symbol should be considered the target of the arc; the operand on the right of the
`\GTSdAD` symbol should be considered the source of the arc.
 $A.\GTSdirectedAdjacencyDown/A.$
or $A.\GTSdirAdjDown/A.$
or $A.\GTSnDAD/A.$
will yield $A.\frac{\rightarrow}{\neg}A.$
`u \GTSdirectedAdjacencyDown/ v`
or `u \GTSdirAdjDown/ v`
or `u \GTSnDAD/ v`
will yield $u \frac{\rightarrow}{\neg} v.$
 $\$u \GTSdirectedAdjacencyDown/ v\$$
or $\$u \GTSdirAdjDown/ v\$$
or $\$u \GTSnDAD/ v\$$
will yield $u \frac{\rightarrow}{\neg} v.$

`\GTSdirectedAdjacencyDown-` This macro will draw a directed adjacency downward symbol with relation spacing in
`Relation` math mode.
`\GTSdirAdjDownRel` $\$u\GTSdirectedAdjacencyDownRelation/v\$$
`\GTSdADR` or $\$u\GTSdirAdjDownRel/v\$$
or $\$u\GTSdADR/v\$$
will yield $u \frac{\rightarrow}{\neg} v.$

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency downward symbol.
`AdjacencyDown` $A.\GTSnegatedDirectedAdjacencyDown/A.$
`\GTSnegDirAdjDown` or $A.\GTSnegDirAdjDown/A.$
`\GTSnDAD` or $A.\GTSnDAD/A.$
will yield $A.\frac{\rightarrow}{\neg}A.$
`u \GTSnegatedDirectedAdjacencyDown/ v`
or `u \GTSnegDirAdjDown/ v`
or `u \GTSnDAD/ v`
will yield $u \frac{\rightarrow}{\neg} v.$
 $\$u \GTSnegatedDirectedAdjacencyDown/ v\$$
or $\$u \GTSnegDirAdjDown/ v\$$
or $\$u \GTSnDAD/ v\$$
will yield $u \frac{\rightarrow}{\neg} v.$

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency downward symbol with relation
`AdjacencyDownRelation` spacing in math mode.
`\GTSnegDirAdjDownRel` $\$u\GTSnegatedDirectedAdjacencyDownRelation/v\$$
`\GTSnDADR`

or $\$u\backslash\text{GTSnegDirAdjDownRel}/v\$$
or $\$u\backslash\text{GTSnDADR}/v\$$
will yield $u \nparallel v$.

\GTS@i This macro will draw any of the incidence symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and **\GTS@j**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

$\{\langle mainColor \rangle\}$ If non-empty, this color will be used as main color.
 $\{\langle secondColor \rangle\}$ If non-empty, this color will be used as second color.
 $\{\langle drawSlash \rangle\}$ Non-empty implies that a negation will be drawn.
 $\{\langle drawUpArrow \rangle\}$ Non-empty implies that an up-arrow will be drawn.
 $\{\langle drawDownArrow \rangle\}$ Non-empty implies that a down-arrow will be drawn.

\GTS@j This macro builds on top of **\GTS@i** to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and **\GTS@i**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**.

Arguments:

$\{\langle mainColor \rangle\}$ If non-empty, this color will be used as main color.
 $\{\langle secondColor \rangle\}$ If non-empty, this color will be used as second color.
 $\{\langle drawSlash \rangle\}$ Non-empty implies that a negation will be drawn.
 $\{\langle drawUpArrow \rangle\}$ Non-empty implies that an up-arrow will be drawn.
 $\{\langle drawDownArrow \rangle\}$ Non-empty implies that a down-arrow will be drawn.
 $\{\langle relation \rangle\}$ Non-empty implies that relation spacing will be added.

\GTSxi Versatile macros for incidence that can be given options.

\GTSxinc Argument:

\GTSxincidence [$\langle optionsList \rangle$] Sets the following options if not empty:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
in or i,
out or o,
relation or rel or r.

\GTSincidence This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

\GTSinc
\GTSi

A.\GTSincidence/A.
or A.\GTSinc/A.
or A.\GTSi/A.
will yield $A \downarrow A$.
v \GTSincidence/ e
or v \GTSinc/ e
or v \GTSi/ e
will yield $v \downarrow e$.
 $\$v \backslash\text{GTSincidence}/ e\$$
or $\$v \backslash\text{GTSinc}/ e\$$
or $\$v \backslash\text{GTSi}/ e\$$
will yield $v \downarrow e$.

\GTSincidenceRelation This macro will draw an incidence symbol with relation spacing in math mode.

\GTSincRel $\$v\backslash\text{GTSincidenceRelation}/e\$$

\GTSiR or $\$v\backslash\text{GTSincRel}/e\$$

or $\$v\backslash\text{GTSiR}/e\$$

will yield $v \downarrow e$.

`\GTSnegatedIncidence` This macro will draw a negated incidence symbol.

`\GTSnegInc` $A.\overline{\text{GTSnegInc}}/A.$

`\GTSnI` or $A.\overline{\text{GTSnegInc}}/A.$

or $A.\overline{\text{GTSnI}}/A.$

will yield $A.\overline{\text{A}}.$

$v \overline{\text{GTSnegatedIncidence}}/e$

or $v \overline{\text{GTSnegInc}}/e$

or $v \overline{\text{GTSnI}}/e$

will yield $v \overline{\text{e}}.$

$\$v \overline{\text{GTSnegatedIncidence}}/e\$$

or $\$v \overline{\text{GTSnegInc}}/e\$$

or $\$v \overline{\text{GTSnI}}/e\$$

will yield $v \overline{\text{e}}.$

`\GTSnegatedIncidence-` This macro will draw a negated incidence symbol with relation spacing in math mode.

`Relation` $\$v \overline{\text{GTSnegatedIncidenceRelation}}/e\$$

`\GTSnegIncRel` or $\$v \overline{\text{GTSnegIncRel}}/e\$$

`\GTSnIR` or $\$v \overline{\text{GTSnIR}}/e\$$

will yield $v \overline{\text{e}}.$

`\GTSoutIncidence` This macro will draw a directed incidence upward (out(ward) incidence) symbol. The

`\GTSoutInc` operand on the left of the symbol should be considered the vertex to which the arc/the

`\GTSoI` operand on the right of the symbol is incident.

$A.\overline{\text{GTSoutIncidence}}/A.$

or $A.\overline{\text{GTSoutInc}}/A.$

or $A.\overline{\text{GTSoI}}/A.$

will yield $A.\overline{\text{A}}.$

$v \overline{\text{GTSoutIncidence}}/e$

or $v \overline{\text{GTSoutInc}}/e$

or $v \overline{\text{GTSoI}}/e$

will yield $v \overline{\text{e}}.$

$\$v \overline{\text{GTSoutIncidence}}/e\$$

or $\$v \overline{\text{GTSoutInc}}/e\$$

or $\$v \overline{\text{GTSoI}}/e\$$

will yield $v \overline{\text{e}}.$

`\GTSoutIncidenceRelation` This macro will draw a directed incidence upward (out(ward) incidence) symbol with

`\GTSoutIncRel` relation spacing in math mode.

`\GTSoIR` $\$v \overline{\text{GTSoutIncidenceRelation}}/e\$$

or $\$v \overline{\text{GTSoutIncRel}}/e\$$

or $\$v \overline{\text{GTSoIR}}/e\$$

will yield $v \overline{\text{e}}.$

`\GTSnegatedOutIncidence` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

`\GTSnegOutInc` $A.\overline{\text{GTSnegatedOutIncidence}}/A.$

`\GTSnOI` or $A.\overline{\text{GTSnegOutInc}}/A.$

or $A.\overline{\text{GTSnOI}}/A.$

will yield $A.\overline{\text{A}}.$

$v \overline{\text{GTSnegatedOutIncidence}}/e$

or $v \overline{\text{GTSnegOutInc}}/e$

or $v \overline{\text{GTSnOI}}/e$

will yield $v \overline{\text{e}}.$

$\$v \overline{\text{GTSnegatedOutIncidence}}/e\$$

or $\$v \overline{\text{GTSnegOutInc}}/e\$$

or $\$v \overline{\text{GTSnOI}}/e\$$

will yield $v \overline{\text{e}}.$

`\GTSnegatedOutIncidence-` This macro will draw a negated directed incidence upward (out(ward) incidence)

`Relation` symbol with relation spacing in math mode.

`\GTSnegOutIncRel` $\$v \overline{\text{GTSnegatedOutIncidenceRelation}}/e\$$

`\GTSnOIR` or $\$v \overline{\text{GTSnegOutIncRel}}/e\$$

or $\$v\backslash\text{GTSnOIR}/e\$$
will yield $v \not\bowtie e$.

$\backslash\text{GTSinIncidence}$ This macro will draw a directed incidence downward (in(ward) incidence) symbol. The
 $\backslash\text{GTSinInc}$ operand on the left of the symbol should be considered the vertex to which the arc/the
 $\backslash\text{GTSiI}$ operand on the right of the symbol is incident.
 $A.\backslash\text{GTSinIncidence}/A.$
or $A.\backslash\text{GTSinInc}/A.$
or $A.\backslash\text{GTSiI}/A.$
will yield $A.\not\bowtie A.$
 $v \backslash\text{GTSinIncidence}/ e$
or $v \backslash\text{GTSinInc}/ e$
or $v \backslash\text{GTSiI}/ e$
will yield $v \not\bowtie e$.
 $\$v \backslash\text{GTSinIncidence}/ e\$$
or $\$v \backslash\text{GTSinInc}/ e\$$
or $\$v \backslash\text{GTSiI}/ e\$$
will yield $v \not\bowtie e$.

$\backslash\text{GTSinIncidenceRelation}$ This macro will draw a directed incidence downward (in(ward) incidence) symbol with
 $\backslash\text{GTSinIncRel}$ relation spacing in math mode.
 $\backslash\text{GTSiIR}$ $\$v\backslash\text{GTSinIncidenceRelation}/e\$$
or $\$v\backslash\text{GTSinIncRel}/e\$$
or $\$v\backslash\text{GTSiIR}/e\$$
will yield $v \not\bowtie e$.

$\backslash\text{GTSnegatedInIncidence}$ This macro will draw a negated directed incidence downward (in(ward) incidence) sym-
 $\backslash\text{GTSnegInInc}$ bol.
 $\backslash\text{GTSnII}$ $A.\backslash\text{GTSnegatedInIncidence}/A.$
or $A.\backslash\text{GTSnegInInc}/A.$
or $A.\backslash\text{GTSnII}/A.$
will yield $A.\not\bowtie A.$
 $v \backslash\text{GTSnegatedInIncidence}/ e$
or $v \backslash\text{GTSnegInInc}/ e$
or $v \backslash\text{GTSnII}/ e$
will yield $v \not\bowtie e$.
 $\$v \backslash\text{GTSnegatedInIncidence}/ e\$$
or $\$v \backslash\text{GTSnegInInc}/ e\$$
or $\$v \backslash\text{GTSnII}/ e\$$
will yield $v \not\bowtie e$.

$\backslash\text{GTSnegatedInIncidence-}$ This macro will draw a negated directed incidence downward (in(ward) incidence)
 Relation symbol with relation spacing in math mode.
 $\backslash\text{GTSnegInIncRel}$ $\$v\backslash\text{GTSnegatedInIncidenceRelation}/e\$$
 $\backslash\text{GTSnIIR}$ or $\$v\backslash\text{GTSnegInIncRel}/e\$$
or $\$v\backslash\text{GTSnIIR}/e\$$
will yield $v \not\bowtie e$.

$\backslash\text{GTSadjacencyMonochrome}$ This macro will draw a monochrome adjacency symbol.
 $\backslash\text{GTSadjMono}$ $A.\backslash\text{GTSadjacencyMonochrome}/A.$
 $\backslash\text{GTSadjMon}$ or $A.\backslash\text{GTSadjMono}/A.$
 $\backslash\text{GTSaM}$ or $A.\backslash\text{GTSadjMon}/A.$
or $A.\backslash\text{GTSaM}/A.$
will yield $A.\not\circ A.$
 $u \backslash\text{GTSadjacencyMonochrome}/ v$
or $u \backslash\text{GTSadjMono}/ v$
or $u \backslash\text{GTSadjMon}/ v$
or $u \backslash\text{GTSaM}/ v$
will yield $u \not\circ v$.
 $\$u \backslash\text{GTSadjacencyMonochrome}/ v\$$
or $\$u \backslash\text{GTSadjMono}/ v\$$

or $\$u \backslash\text{GTSadjMon}/ v\$$
or $\$u \backslash\text{GTSaM}/ v\$$
will yield $u \circ v$.

$\backslash\text{GTSadjacencyMonochrome-}$ This macro will draw a monochrome adjacency symbol with relation spacing in math
Relation mode.

$\backslash\text{GTSadjMonoRel}$ $\$u \backslash\text{GTSadjacencyMonochromeRelation}/ v\$$
 $\backslash\text{GTSadjMonRel}$ or $\$u \backslash\text{GTSadjMonoRel}/ v\$$
 $\backslash\text{GTSaMR}$ or $\$u \backslash\text{GTSadjMonRel}/ v\$$
or $\$u \backslash\text{GTSaMR}/ v\$$
will yield $u \circ v$.

$\backslash\text{GTSunadjacencyMonochrome}$ This macro will draw a monochrome unadjacency symbol.

$\backslash\text{GTSunadjMono}$ $A. \backslash\text{GTSunadjacencyMonochrome}/ A.$
 $\backslash\text{GTSunadjMon}$ or $A. \backslash\text{GTSunadjMono}/ A.$
 $\backslash\text{GTSuM}$ or $A. \backslash\text{GTSunadjMon}/ A.$
or $A. \backslash\text{GTSuM}/ A.$
will yield $A \circ A$.
 $u \backslash\text{GTSunadjacencyMonochrome}/ v$
or $u \backslash\text{GTSunadjMono}/ v$
or $u \backslash\text{GTSunadjMon}/ v$
or $u \backslash\text{GTSuM}/ v$
will yield $u \circ v$.
 $\$u \backslash\text{GTSunadjacencyMonochrome}/ v\$$
or $\$u \backslash\text{GTSunadjMono}/ v\$$
or $\$u \backslash\text{GTSunadjMon}/ v\$$
or $\$u \backslash\text{GTSuM}/ v\$$
will yield $u \circ v$.

$\backslash\text{GTSunadjacencyMonochrome-}$ This macro will draw a monochrome unadjacency symbol with relation spacing in
Relation math mode.

$\backslash\text{GTSunadjMonoRel}$ $\$u \backslash\text{GTSunadjacencyMonochromeRelation}/ v\$$
 $\backslash\text{GTSunadjMonRel}$ or $\$u \backslash\text{GTSunadjMonoRel}/ v\$$
 $\backslash\text{GTSuMR}$ or $\$u \backslash\text{GTSunadjMonRel}/ v\$$
or $\$u \backslash\text{GTSuMR}/ v\$$
will yield $u \circ v$.

$\backslash\text{GTSdirectedAdjacencyUp-}$ This macro will draw a monochrome directed adjacency upward symbol. The operand
Monochrome on the left of the symbol should be considered the source of the arc; the operand on the
right of the symbol should be considered the target of the arc.

$\backslash\text{GTSdirAdjUpMono}$ $A. \backslash\text{GTSdirectedAdjacencyUpMonochrome}/ A.$
 $\backslash\text{GTSdirAdjUpMon}$ or $A. \backslash\text{GTSdirAdjUpMono}/ A.$
or $A. \backslash\text{GTSdirAdjUpMon}/ A.$
or $A. \backslash\text{GTSdAUM}/ A.$
will yield $A \circ A$.
 $u \backslash\text{GTSdirectedAdjacencyUpMonochrome}/ v$
or $u \backslash\text{GTSdirAdjUpMono}/ v$
or $u \backslash\text{GTSdirAdjUpMon}/ v$
or $u \backslash\text{GTSdAUM}/ v$
will yield $u \circ v$.
 $\$u \backslash\text{GTSdirectedAdjacencyUpMonochrome}/ v\$$
or $\$u \backslash\text{GTSdirAdjUpMono}/ v\$$
or $\$u \backslash\text{GTSdirAdjUpMon}/ v\$$
or $\$u \backslash\text{GTSdAUM}/ v\$$
will yield $u \circ v$.

$\backslash\text{GTSdirectedAdjacencyUp-}$ This macro will draw a monochrome directed adjacency upward symbol with relation
MonochromeRelation spacing in math mode.

$\backslash\text{GTSdirAdjUpMonoRel}$ $\$u \backslash\text{GTSdirectedAdjacencyUpMonochromeRelation}/ v\$$
 $\backslash\text{GTSdirAdjUpMonRel}$ or $\$u \backslash\text{GTSdirAdjUpMonoRel}/ v\$$
 $\backslash\text{GTSdAUMR}$ or $\$u \backslash\text{GTSdirAdjUpMonRel}/ v\$$

or $\$u\backslash\text{GTSdAUMR}/v\$$
will yield $u \overset{\circ}{\circ} v$.

$\backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}$ This macro will draw a monochrome negated directed adjacency upward symbol.
A. $\backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/A$.
 $\backslash\text{GTSnegDirAdjUpMono}$ or A. $\backslash\text{GTSnegDirAdjUpMono}/A$.
 $\backslash\text{GTSnegDirAdjUpMon}$ or A. $\backslash\text{GTSnegDirAdjUpMon}/A$.
 $\backslash\text{GTSnDAUM}$ or A. $\backslash\text{GTSnDAUM}/A$.
will yield $A \overset{\circ}{\circ} A$.
u $\backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/ v$
or u $\backslash\text{GTSnegDirAdjUpMono}/ v$
or u $\backslash\text{GTSnegDirAdjUpMon}/ v$
or u $\backslash\text{GTSnDAUM}/ v$
will yield $u \overset{\circ}{\circ} v$.
 $\$u \backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/ v\$$
or $\$u \backslash\text{GTSnegDirAdjUpMono}/ v\$$
or $\$u \backslash\text{GTSnegDirAdjUpMon}/ v\$$
or $\$u \backslash\text{GTSnDAUM}/ v\$$
will yield $u \overset{\circ}{\circ} v$.

$\backslash\text{GTSnegatedDirectedAdjacencyUpMonochromeRelation}$ This macro will draw a monochrome negated directed adjacency upward symbol with
relation spacing in math mode.
 $\$u\backslash\text{GTSnegatedDirectedAdjacencyUpMonochromeRelation}/v\$$
 $\backslash\text{GTSnegDirAdjUpMonoRel}$ or $\$u\backslash\text{GTSnegDirAdjUpMonoRel}/v\$$
 $\backslash\text{GTSnegDirAdjUpMonRel}$ or $\$u\backslash\text{GTSnegDirAdjUpMonRel}/v\$$
 $\backslash\text{GTSnDAUMR}$ or $\$u\backslash\text{GTSnDAUMR}/v\$$
will yield $u \overset{\circ}{\circ} v$.

$\backslash\text{GTSdirectedAdjacencyDownMonochrome}$ This macro will draw a monochrome directed adjacency downward symbol. The
operand on the left of the symbol should be considered the target of the arc; the operand
 $\backslash\text{GTSdirAdjDownMono}$ on the right of the symbol should be considered the source of the arc.
 $\backslash\text{GTSdirAdjDownMon}$ A. $\backslash\text{GTSdirectedAdjacencyDownMonochrome}/A$.
 $\backslash\text{GTSdADM}$ or A. $\backslash\text{GTSdirAdjDownMono}/A$.
or A. $\backslash\text{GTSdirAdjDownMon}/A$.
or A. $\backslash\text{GTSdADM}/A$.
will yield $A \overset{\circ}{\circ} A$.
u $\backslash\text{GTSdirectedAdjacencyDownMonochrome}/ v$
or u $\backslash\text{GTSdirAdjDownMono}/ v$
or u $\backslash\text{GTSdirAdjDownMon}/ v$
or u $\backslash\text{GTSdADM}/ v$
will yield $u \overset{\circ}{\circ} v$.
 $\$u \backslash\text{GTSdirectedAdjacencyDownMonochrome}/ v\$$
or $\$u \backslash\text{GTSdirAdjDownMono}/ v\$$
or $\$u \backslash\text{GTSdirAdjDownMon}/ v\$$
or $\$u \backslash\text{GTSdADM}/ v\$$
will yield $u \overset{\circ}{\circ} v$.

$\backslash\text{GTSdirectedAdjacencyDownMonochromeRelation}$ This macro will draw a monochrome directed adjacency downward symbol with
relation spacing in math mode.
 $\$u\backslash\text{GTSdirectedAdjacencyDownMonochromeRelation}/v\$$
 $\backslash\text{GTSdirAdjDownMonoRel}$ or $\$u\backslash\text{GTSdirAdjDownMonoRel}/v\$$
 $\backslash\text{GTSdirAdjDownMonRel}$ or $\$u\backslash\text{GTSdirAdjDownMonRel}/v\$$
 $\backslash\text{GTSdADMR}$ or $\$u\backslash\text{GTSdADMR}/v\$$
or $\$u\backslash\text{GTSdADMR}/v\$$
will yield $u \overset{\circ}{\circ} v$.

$\backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}$ This macro will draw a monochrome negated directed adjacency downward symbol.
A. $\backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/A$.
 $\backslash\text{GTSnegDirAdjDownMono}$ or A. $\backslash\text{GTSnegDirAdjDownMono}/A$.
 $\backslash\text{GTSnegDirAdjDownMon}$ or A. $\backslash\text{GTSnegDirAdjDownMon}/A$.
 $\backslash\text{GTSnDADM}$ or A. $\backslash\text{GTSnDADM}/A$.
will yield $A \overset{\circ}{\circ} A$.

$\backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/ v$
or $\backslash\text{GTSnegDirAdjDownMono}/ v$
or $\backslash\text{GTSnegDirAdjDownMon}/ v$
or $\backslash\text{GTSnDADM}/ v$
will yield $u \overset{\circ}{\rightharpoonup} v$.
 $\$u \backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/ v\$$
or $\$u \backslash\text{GTSnegDirAdjDownMono}/ v\$$
or $\$u \backslash\text{GTSnegDirAdjDownMon}/ v\$$
or $\$u \backslash\text{GTSnDADM}/ v\$$
will yield $u \overset{\circ}{\rightharpoonup} v$.

$\backslash\text{GTSnegatedDirectedAdjacencyDownMonochromeRelation}$ This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.
 $\backslash\text{GTSnegDirAdjDownMonoRel}$ $\$u \backslash\text{GTSnegatedDirectedAdjacencyDownMonochromeRelation}/ v\$$
 $\backslash\text{GTSnegDirAdjDownMonRel}$ or $\$u \backslash\text{GTSnegDirAdjDownMonoRel}/ v\$$
 $\backslash\text{GTSnDADMR}$ or $\$u \backslash\text{GTSnDADMR}/ v\$$
will yield $u \overset{\circ}{\rightharpoonup} v$.

$\backslash\text{GTSincidenceMonochrome}$ This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.
 $\backslash\text{GTSiM}$ $A. \backslash\text{GTSincidenceMonochrome}/ A.$
or $A. \backslash\text{GTSincMono}/ A.$
or $A. \backslash\text{GTSincMon}/ A.$
or $A. \backslash\text{GTSiM}/ A.$
will yield $A. \downarrow A.$
 $v \backslash\text{GTSincidenceMonochrome}/ e$
or $v \backslash\text{GTSincMono}/ e$
or $v \backslash\text{GTSincMon}/ e$
or $v \backslash\text{GTSiM}/ e$
will yield $v \downarrow e$.
 $\$v \backslash\text{GTSincidenceMonochrome}/ e\$$
or $\$v \backslash\text{GTSincMono}/ e\$$
or $\$v \backslash\text{GTSincMon}/ e\$$
or $\$v \backslash\text{GTSiM}/ e\$$
will yield $v \downarrow e$.

$\backslash\text{GTSincidenceMonochromeRelation}$ This macro will draw a monochrome incidence symbol with relation spacing in math mode.
 $\backslash\text{GTSincMonoRel}$ $\$v \backslash\text{GTSincidenceMonochromeRelation}/ e\$$
 $\backslash\text{GTSincMonRel}$ or $\$v \backslash\text{GTSincMonoRel}/ e\$$
 $\backslash\text{GTSiMR}$ or $\$v \backslash\text{GTSincMonRel}/ e\$$
or $\$v \backslash\text{GTSiMR}/ e\$$
will yield $v \downarrow e$.

$\backslash\text{GTSnegatedIncidenceMonochrome}$ This macro will draw a monochrome negated incidence symbol.
 $\backslash\text{GTSnegIncMono}$ $A. \backslash\text{GTSnegatedIncidenceMonochrome}/ A.$
 $\backslash\text{GTSnegIncMon}$ or $A. \backslash\text{GTSnegIncMono}/ A.$
 $\backslash\text{GTSnIM}$ or $A. \backslash\text{GTSnIM}/ A.$
will yield $A. \nmid A.$
 $v \backslash\text{GTSnegatedIncidenceMonochrome}/ e$
or $v \backslash\text{GTSnegIncMono}/ e$
or $v \backslash\text{GTSnegIncMon}/ e$
or $v \backslash\text{GTSnIM}/ e$
will yield $v \nmid e$.
 $\$v \backslash\text{GTSnegatedIncidenceMonochrome}/ e\$$
or $\$v \backslash\text{GTSnegIncMono}/ e\$$
or $\$v \backslash\text{GTSnegIncMon}/ e\$$
or $\$v \backslash\text{GTSnIM}/ e\$$

will yield $v \not\hookrightarrow e$.

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol with relation spacing
`MonochromeRelation` in math mode.
`\GTSnegIncMonoRel` $\$v\backslash\text{GTSnegatedIncidenceMonochromeRelation}/e\$$
`\GTSnegIncMonRel` or $\$v\backslash\text{GTSnegIncMonoRel}/e\$$
`\GTSnIMR` or $\$v\backslash\text{GTSnegIncMonRel}/e\$$
or $\$v\backslash\text{GTSnIMR}/e\$$
will yield $v \not\hookrightarrow e$.

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence)
`\GTSoutIncMono` symbol. The operand on the left of the symbol should be considered the vertex to which
`\GTSoutIncMon` the arc/the operand on the right of the symbol is incident.
`\GTSoIM` $A.\backslash\text{GTSoutIncidenceMonochrome}/A.$
or $A.\backslash\text{GTSoutIncMono}/A.$
or $A.\backslash\text{GTSoutIncMon}/A.$
or $A.\backslash\text{GTSoIM}/A.$
will yield $A.\not\hookrightarrow A.$
 $v \backslash\text{GTSoutIncidenceMonochrome}/ e$
or $v \backslash\text{GTSoutIncMono}/ e$
or $v \backslash\text{GTSoutIncMon}/ e$
or $v \backslash\text{GTSoIM}/ e$
will yield $v \not\hookrightarrow e$.
 $\$v \backslash\text{GTSoutIncidenceMonochrome}/ e\$$
or $\$v \backslash\text{GTSoutIncMono}/ e\$$
or $\$v \backslash\text{GTSoutIncMon}/ e\$$
or $\$v \backslash\text{GTSoIM}/ e\$$
will yield $v \not\hookrightarrow e$.

`\GTSoutIncidenceMonochrome-` This macro will draw a monochrome directed incidence upward (out(ward) incidence)
`Relation` symbol with relation spacing in math mode.
`\GTSoutIncMonoRel` $\$v\backslash\text{GTSoutIncidenceMonochromeRelation}/e\$$
`\GTSoutIncMonRel` or $\$v\backslash\text{GTSoutIncMonoRel}/e\$$
`\GTSoIMR` or $\$v\backslash\text{GTSoutIncMonRel}/e\$$
or $\$v\backslash\text{GTSoIMR}/e\$$
will yield $v \not\hookrightarrow e$.

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward)
`Monochrome` incidence) symbol.
`\GTSnegOutIncMono` $A.\backslash\text{GTSnegatedOutIncidenceMonochrome}/A.$
`\GTSnegOutIncMon` or $A.\backslash\text{GTSnegOutIncMono}/A.$
`\GTSnOIM` or $A.\backslash\text{GTSnegOutIncMon}/A.$
or $A.\backslash\text{GTSnOIM}/A.$
will yield $A.\not\hookrightarrow A.$
 $v \backslash\text{GTSnegatedOutIncidenceMonochrome}/ e$
or $v \backslash\text{GTSnegOutIncMono}/ e$
or $v \backslash\text{GTSnegOutIncMon}/ e$
or $v \backslash\text{GTSnOIM}/ e$
will yield $v \not\hookrightarrow e$.
 $\$v \backslash\text{GTSnegatedOutIncidenceMonochrome}/ e\$$
or $\$v \backslash\text{GTSnegOutIncMono}/ e\$$
or $\$v \backslash\text{GTSnegOutIncMon}/ e\$$
or $\$v \backslash\text{GTSnOIM}/ e\$$
will yield $v \not\hookrightarrow e$.

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward)
`MonochromeRelation` incidence) symbol with relation spacing in math mode.
`\GTSnegOutIncMonoRel` $\$v\backslash\text{GTSnegatedOutIncidenceMonochromeRelation}/e\$$
`\GTSnegOutIncMonRel` or $\$v\backslash\text{GTSnegOutIncMonoRel}/e\$$
`\GTSnOIMR` or $\$v\backslash\text{GTSnegOutIncMonRel}/e\$$
or $\$v\backslash\text{GTSnOIMR}/e\$$

will yield $v \nrightarrow e$.

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which
`\GTSinIncMono` the arc/the operand on the right of the symbol is incident.
`\GTSiIM` `A.\GTSinIncidenceMonochrome/A.`
or `A.\GTSinIncMono/A.`
or `A.\GTSinIncMon/A.`
or `A.\GTSiIM/A.`
will yield $A \nrightarrow A$.
`v \GTSinIncidenceMonochrome/ e`
or `v \GTSinIncMono/ e`
or `v \GTSinIncMon/ e`
or `v \GTSiIM/ e`
will yield $v \nrightarrow e$.
`$v \GTSinIncidenceMonochrome/ e$`
or `$v \GTSinIncMono/ e$`
or `$v \GTSinIncMon/ e$`
or `$v \GTSiIM/ e$`
will yield $v \nrightarrow e$.

`\GTSinIncidenceMonochrome-` This macro will draw a monochrome directed incidence downward (in(ward) inci-
`Relation` dence) symbol with relation spacing in math mode.
`\GTSinIncMonoRel` `$v\GTSinIncidenceMonochromeRelation/e$`
`\GTSinIncMonRel` or `$v\GTSinIncMonoRel/e$`
`\GTSiIMR` or `$v\GTSinIncMonRel/e$`
or `$v\GTSiIMR/e$`
will yield $v \nrightarrow e$.

`\GTSnegatedInIncidence-` This macro will draw a monochrome negated directed incidence downward (in(ward)
`Monochrome` incidence) symbol.
`\GTSnegInIncMono` `A.\GTSnegatedInIncidenceMonochrome/A.`
`\GTSnegInIncMon` or `A.\GTSnegInIncMono/A.`
`\GTSnIIM` or `A.\GTSnegInIncMon/A.`
or `A.\GTSnIIM/A.`
will yield $A \nrightarrow A$.
`v \GTSnegatedInIncidenceMonochrome/ e`
or `v \GTSnegInIncMono/ e`
or `v \GTSnegInIncMon/ e`
or `v \GTSnIIM/ e`
will yield $v \nrightarrow e$.
`$v \GTSnegatedInIncidenceMonochrome/ e$`
or `$v \GTSnegInIncMono/ e$`
or `$v \GTSnegInIncMon/ e$`
or `$v \GTSnIIM/ e$`
will yield $v \nrightarrow e$.

`\GTSnegatedInIncidence-` This macro will draw a monochrome negated directed incidence downward (in(ward)
`MonochromeRelation` incidence) symbol with relation spacing in math mode.
`\GTSnegInIncMonoRel` `$v\GTSnegatedInIncidenceMonochromeRelation/e$`
`\GTSnegInIncMonRel` or `$v\GTSnegInIncMonoRel/e$`
`\GTSnIIMR` or `$v\GTSnegInIncMonRel/e$`
or `$v\GTSnIIMR/e$`
will yield $v \nrightarrow e$.

3 Implementation

```
1 \package
2 \RequirePackage{amsmath}
```

```

3 \RequirePackage{amssymb}
4 \RequirePackage{graphicx}
5 \RequirePackage{keyval}
6 \RequirePackage{fdsymbol}
7 \RequirePackage{lua-unicode-math}
8 \RequirePackage{xcolor}
9
10 \def\GTS@mainColor{black}
11 \def\GTS@secondColor{gray}
12 \def\GTS@mainColorBackup{black}
13 \def\GTS@secondColorBackup{gray}
14 \def\GTS@drawSlash{}
15 \def\GTS@drawUpArrow{}
16 \def\GTS@drawDownArrow{}
17

```

\GTS@newcommand This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

$\langle commandname \rangle$ The name of the command to create.

$\langle commandcode \rangle$ The code of the command to create.

```

18 \@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
19   \@ifdefinable{#1}{\def#1/{#2}}
20 }}

```

\GTSsetMainColor This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

$\langle color \rangle$ The color that must be used by GraphTheorySymbol afterward.

```

21 \newcommand{\GTSsetMainColor}[1]{%
22 \edef\GTS@mainColorTemp{#1}%
23 \edef\GTS@mainColorBackup{\GTS@mainColor}%
24 \edef\GTS@mainColor{\GTS@mainColorTemp}%
25 }

```

\GTSsetSecondColor This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

$\langle color \rangle$ The color that must be used by GraphTheorySymbol afterward.

```

26 \newcommand{\GTSsetSecondColor}[1]{%
27 \edef\GTS@secondColorTemp{#1}%
28 \edef\GTS@secondColorBackup{\GTS@secondColor}%
29 \edef\GTS@secondColor{\GTS@secondColorTemp}%
30 }

```

\GTS@a This macro will draw any adjacency symbol given the correct parameters. Its use is purely technical. See the macros derived from it and **\GTS@b**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

$\langle mainColor \rangle$ If non-empty, this color will be used as main color.

$\langle secondColor \rangle$ If non-empty, this color will be used as second color.

$\langle drawSlash \rangle$ Non-empty implies that a negation will be drawn.

$\langle drawUpArrow \rangle$ Non-empty implies that an up-arrow will be drawn.

$\langle drawDownArrow \rangle$ Non-empty implies that a down-arrow will be drawn.

```

31 \newcommand{\GTS@a}[5]{%
32 \if#1\empty\else%
33 \GTSsetMainColor{#1}%
34 \fi%
35 \if#2\empty\else%
36 \GTSsetSecondColor{#2}%
37 \fi%
38 \text{\ensuremath{%
39 \raisebox{-0.2ex}{\scalebox{1.5}[1.5]{%
40 %

```

```

41 \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
42 $\mathcolor{\GTS@secondColor}{\circ}$%
43 }}%
44 \hspace{-0.39ex}%
45 \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
46 $\mathcolor{\GTS@secondColor}{\bullet}$%
47 }}%
48 \hspace{-0.12ex}%
49 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
50 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
51 }}%
52 \hspace{-0.40ex}%
53 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
54 $\mathcolor{\GTS@secondColor}{\circ}$%
55 }}%
56 \hspace{-0.39ex}%
57 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
58 $\mathcolor{\GTS@secondColor}{\bullet}$%
59 }}%
60 \hspace{0.27ex}%
if GTS@drawUpArrow empty else
61 \if#4\empty\else%
62 \hspace{-0.65ex}%
63 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
64 \textcolor{\GTS@mainColor}{\textasciicircum}%
65 }}%
66 \fi%
if GTS@drawDownArrow empty else
67 \if#5\empty\else%
68 \hspace{-0.65ex}%
69 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
70 \textcolor{\GTS@mainColor}{\textasciicircum}%
71 }}%
72 \fi%
if GTS@drawSlash empty else
73 \if#3\empty\else%
74 \hspace{-0.65ex}%
75 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
76 \textcolor{black}{/}%
77 }}%
78 \fi%
79 %
80 }}%
81 }}%
82 \if#1\empty\else%
83 \GTSsetMainColor{\GTS@mainColorBackup}%
84 \fi%
85 \if#2\empty\else%
86 \GTSsetSecondColor{\GTS@secondColorBackup}%
87 \fi%
88 }

```

\GTS@b This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and **\GTS@a**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

- {<mainColor>}** If non-empty, this color will be used as main color.
- {<secondColor>}** If non-empty, this color will be used as second color.
- {<drawSlash>}** Non-empty implies that a negation will be drawn.
- {<drawUpArrow>}** Non-empty implies that an up-arrow will be drawn.
- {<drawDownArrow>}** Non-empty implies that a down-arrow will be drawn.
- {<relation>}** Non-empty implies that relation spacing will be added.

```

89 \newcommand{\GTS@b}[6]{%
90 \if#61%

```



```

91 \mathrel{\GTS@a{#1}{#2}{#3}{#4}{#5}}%
92 \else%
93 \GTS@a{#1}{#2}{#3}{#4}{#5}%
94 \fi%
95 }

```

`\GTSxa` Versatile macros for adjacency that can be given options.

`\GTSxadj` Argument:

`\GTSxadjacency` [*optionsList*] Sets the following options if not empty:

mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
up or u,
down or d,
relation or rel or r.

```

96 \define@key{GTSxa}{mainColor}{\def\GTS@@mainColor{#1}}
97 \define@key{GTSxa}{mainC}{\def\GTS@@mainColor{#1}}
98 \define@key{GTSxa}{mColor}{\def\GTS@@mainColor{#1}}
99 \define@key{GTSxa}{mC}{\def\GTS@@mainColor{#1}}
100 \define@key{GTSxa}{secondColor}{\def\GTS@@secondColor{#1}}
101 \define@key{GTSxa}{secondC}{\def\GTS@@secondColor{#1}}
102 \define@key{GTSxa}{sColor}{\def\GTS@@secondColor{#1}}
103 \define@key{GTSxa}{sC}{\def\GTS@@secondColor{#1}}
104 \define@key{GTSxa}{monochrome}[1]%
105 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
106 \define@key{GTSxa}{mono}[1]%
107 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
108 \define@key{GTSxa}{mon}[1]%
109 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
110 \define@key{GTSxa}{m}[1]%
111 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
112 \define@key{GTSxa}{negated}[1]{\def\GTS@@drawSlash{1}}
113 \define@key{GTSxa}{not}[1]{\def\GTS@@drawSlash{1}}
114 \define@key{GTSxa}{n}[1]{\def\GTS@@drawSlash{1}}
115 \define@key{GTSxa}{up}[1]{\def\GTS@@drawUpArrow{1}}
116 \define@key{GTSxa}{u}[1]{\def\GTS@@drawUpArrow{1}}
117 \define@key{GTSxa}{down}[1]{\def\GTS@@drawDownArrow{1}}
118 \define@key{GTSxa}{d}[1]{\def\GTS@@drawDownArrow{1}}
119 \define@key{GTSxa}{relation}[1]{\def\GTS@@relation{1}}
120 \define@key{GTSxa}{rel}[1]{\def\GTS@@relation{1}}
121 \define@key{GTSxa}{r}[1]{\def\GTS@@relation{1}}
122
123 \newcommand{\GTSxa}[1] [] {%
124 \def\GTS@@mainColor{}%
125 \def\GTS@@secondColor{}%
126 \def\GTS@@drawSlash{}%
127 \def\GTS@@drawUpArrow{}%
128 \def\GTS@@drawDownArrow{}%
129 \def\GTS@@relation{}%
130 \setkeys{GTSxa}{#1}%
131 \GTS@b{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
132 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
133 }
134
135 \newcommand{\GTSxadj}[1] [] {\GTSxa[#1]}
136
137 \newcommand{\GTSxadjacency}[1] [] {\GTSxa[#1]}

```

`\GTSadjacency` This macro will draw an adjacency symbol.

`\GTSadj` 138 `\GTS@newcommand{\GTSadjacency}{\GTS@a{}{}{}{}}{}`

`\GTSa` 139 `\GTS@newcommand{\GTSadj}{\GTS@a{}{}{}{}}{}`

140 `\GTS@newcommand{\GTSa}{\GTS@a{}{}{}{}}{}`

`\GTSadjacencyRelation` This macro will draw an adjacency symbol with relation spacing in math mode.

`\GTSadjRel`

`\GTSaR`

```

141 \GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSa/}}
142 \GTS@newcommand{\GTSadjRel}{\mathrel{\GTSa/}}
143 \GTS@newcommand{\GTSaR}{\mathrel{\GTSa/}}

```

`\GTSunadjacency` This macro will draw an unadjacency symbol.

```

\GTSunadj 144 \GTS@newcommand{\GTSunadjacency}{\GTS@a{}{}{1}{}}
\GTSu 145 \GTS@newcommand{\GTSunadj}{\GTS@a{}{}{1}{}}
146 \GTS@newcommand{\GTSu}{\GTS@a{}{}{1}{}}

```

`\GTSunadjacencyRelation` This macro will draw an unadjacency symbol with relation spacing in math mode.

```

\GTSunadjRel 147 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSu/}}
\GTSuR 148 \GTS@newcommand{\GTSunadjRel}{\mathrel{\GTSu/}}
149 \GTS@newcommand{\GTSuR}{\mathrel{\GTSu/}}

```

`\GTSdirectedAdjacencyUp` This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```

150 \GTS@newcommand{\GTSdirectedAdjacencyUp}{\GTS@a{}{}{1}{}}
151 \GTS@newcommand{\GTSdirAdjUp}{\GTS@a{}{}{1}{}}
152 \GTS@newcommand{\GTSdAU}{\GTS@a{}{}{1}{}}

```

`\GTSdirectedAdjacencyUpRelation` This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

```

\GTSdirAdjUpRel 153 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{\mathrel{\GTSdAU/}}
\GTSdAUR 154 \GTS@newcommand{\GTSdirAdjUpRel}{\mathrel{\GTSdAU/}}
155 \GTS@newcommand{\GTSdAUR}{\mathrel{\GTSdAU/}}

```

`\GTSnegatedDirectedAdjacencyUp` This macro will draw a negated directed adjacency upward symbol.

```

156 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{\GTS@a{}{}{1}{1}{}}
\GTSnegDirAdjUp 157 \GTS@newcommand{\GTSnegDirAdjUp}{\GTS@a{}{}{1}{1}{}}
\GTSnDAU 158 \GTS@newcommand{\GTSnDAU}{\GTS@a{}{}{1}{1}{}}

```

`\GTSnegatedDirectedAdjacencyUpRelation` This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

```

\GTSnegDirAdjUpRel 159 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
\GTSnDAUR 160 \mathrel{\GTSnDAU/}%
161 }
162 \GTS@newcommand{\GTSnegDirAdjUpRel}{\mathrel{\GTSnDAU/}}
163 \GTS@newcommand{\GTSnDAUR}{\mathrel{\GTSnDAU/}}

```

`\GTSdirectedAdjacencyDown` This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

164 \GTS@newcommand{\GTSdirectedAdjacencyDown}{\GTS@a{}{}{}{1}}
165 \GTS@newcommand{\GTSdirAdjDown}{\GTS@a{}{}{}{1}}
166 \GTS@newcommand{\GTSdAD}{\GTS@a{}{}{}{1}}

```

`\GTSdirectedAdjacencyDownRelation` This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

```

\GTSdirAdjDownRel 167 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{\mathrel{\GTSdAD/}}
\GTSdADR 168 \GTS@newcommand{\GTSdirAdjDownRel}{\mathrel{\GTSdAD/}}
169 \GTS@newcommand{\GTSdADR}{\mathrel{\GTSdAD/}}

```

`\GTSnegatedDirectedAdjacencyDown` This macro will draw a negated directed adjacency downward symbol.

```

170 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{\GTS@a{}{}{1}{1}{}}
\GTSnegDirAdjDown 171 \GTS@newcommand{\GTSnegDirAdjDown}{\GTS@a{}{}{1}{1}{}}
\GTSnDAD 172 \GTS@newcommand{\GTSnDAD}{\GTS@a{}{}{1}{1}{}}

```

`\GTSnegatedDirectedAdjacencyDownRelation` This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

```

\GTSnegDirAdjDownRel 173 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
\GTSnDADR 174 \mathrel{\GTSnDAD/}%
175 }
176 \GTS@newcommand{\GTSnegDirAdjDownRel}{\mathrel{\GTSnDAD/}}
177 \GTS@newcommand{\GTSnDADR}{\mathrel{\GTSnDAD/}}

```

`\GTS@i` This macro will draw any incidence symbol given the correct parameters. Its use is purely technical. See the macros derived from it and `\GTS@j`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

`{\mainColor}` If non-empty, this color will be used as main color.

`{\secondColor}` If non-empty, this color will be used as second color.

`{\drawSlash}` Non-empty implies that a negation will be drawn.

`{\drawUpArrow}` Non-empty implies that an up-arrow will be drawn.

`{\drawDownArrow}` Non-empty implies that a down-arrow will be drawn.

```

178 \newcommand{\GTS@i}[5]{%
179 \if#1\empty\else%
180 \GTSsetMainColor{#1}%
181 \fi%
182 \if#2\empty\else%
183 \GTSsetSecondColor{#2}%
184 \fi%
185 \text{\ensuremath{%
186 \raisebox{0ex}{\scalebox{1.8}[1.8]{%
187 %
188 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
189 $\mathcolor{\GTS@secondColor}{\circ}$%
190 }}%
191 \hspace{-0.39ex}%
192 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
193 $\mathcolor{\GTS@secondColor}{\bullet}$%
194 }}%
195 \hspace{-0.12ex}%
196 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
197 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
198 }}%
199 \hspace{-0.18ex}%
200 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
201 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
202 }}%
203 \hspace{0.2ex}%
if GTS@drawUpArrow empty else
204 \if#4\empty\else%
205 \hspace{-0.65ex}%
206 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
207 \textcolor{\GTS@secondColor}{\textasciicircum}%
208 }}%
209 \fi%
if GTS@drawDownArrow empty else
210 \if#5\empty\else%
211 \hspace{-0.65ex}%
212 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
213 \textcolor{\GTS@secondColor}{\textasciicircum}%
214 }}%
215 \fi%
if GTS@drawSlash empty else
216 \if#3\empty\else%
217 \hspace{-0.65ex}%
218 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
219 \textcolor{black}{/}%
220 }}%
221 \fi%
222 %
223 }}%
224 }}%
225 \if#1\empty\else%
226 \GTSsetMainColor{\GTS@mainColorBackup}%
227 \fi%
```

```

228 \if#2\empty\else%
229 \GTSsetSecondColor{\GTS@secondColorBackup}%
230 \fi%
231 }

```

\GTS@j This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and **\GTS@i**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**.

Arguments:

{\mainColor} If non-empty, this color will be used as main color.
{\secondColor} If non-empty, this color will be used as second color.
{\drawSlash} Non-empty implies that a negation will be drawn.
{\drawUpArrow} Non-empty implies that an up-arrow will be drawn.
{\drawDownArrow} Non-empty implies that a down-arrow will be drawn.
{\relation} Non-empty implies that relation spacing will be added.

```

232 \newcommand{\GTS@j}[6]{%
233 \if#61%
234 \mathrel{\GTS@i{#1}{#2}{#3}{#4}{#5}}%
235 \else%
236 \GTS@i{#1}{#2}{#3}{#4}{#5}%
237 \fi%
238 }

```

\GTSxi Versatile macros for incidence that can be given options.

\GTSxinc Argument:

\GTSxincidence [*optionsList*] Sets the following options if not empty:
 mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
 secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
 monochrome or mono or mon or m (equivalent to mC=black,sC=black),
 negated or not or n,
 in or i,
 out or o,
 relation or rel or r.

```

239 \define@key{GTSxi}{mainColor}{\def\GTS@@mainColor{#1}}
240 \define@key{GTSxi}{mainC}{\def\GTS@@mainColor{#1}}
241 \define@key{GTSxi}{mColor}{\def\GTS@@mainColor{#1}}
242 \define@key{GTSxi}{mC}{\def\GTS@@mainColor{#1}}
243 \define@key{GTSxi}{secondColor}{\def\GTS@@secondColor{#1}}
244 \define@key{GTSxi}{secondC}{\def\GTS@@secondColor{#1}}
245 \define@key{GTSxi}{sColor}{\def\GTS@@secondColor{#1}}
246 \define@key{GTSxi}{sC}{\def\GTS@@secondColor{#1}}
247 \define@key{GTSxi}{monochrome}[1]%
248 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
249 \define@key{GTSxi}{mono}[1]%
250 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
251 \define@key{GTSxi}{mon}[1]%
252 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
253 \define@key{GTSxi}{m}[1]%
254 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
255 \define@key{GTSxi}{negated}[1]{\def\GTS@@drawSlash{1}}
256 \define@key{GTSxi}{not}[1]{\def\GTS@@drawSlash{1}}
257 \define@key{GTSxi}{n}[1]{\def\GTS@@drawSlash{1}}
258 \define@key{GTSxi}{out}[1]{\def\GTS@@drawUpArrow{1}}
259 \define@key{GTSxi}{o}[1]{\def\GTS@@drawUpArrow{1}}
260 \define@key{GTSxi}{in}[1]{\def\GTS@@drawDownArrow{1}}
261 \define@key{GTSxi}{i}[1]{\def\GTS@@drawDownArrow{1}}
262 \define@key{GTSxi}{relation}[1]{\def\GTS@@relation{1}}
263 \define@key{GTSxi}{rel}[1]{\def\GTS@@relation{1}}
264 \define@key{GTSxi}{r}[1]{\def\GTS@@relation{1}}
265
266 \newcommand{\GTSxi}[1][]{%
267 \def\GTS@@mainColor{}%
268 \def\GTS@@secondColor{}%
269 \def\GTS@@drawSlash{}%

```

```

270 \def\GTS@@drawUpArrow{%
271 \def\GTS@@drawDownArrow{%
272 \def\GTS@@relation{%
273 \setkeys{GTSxi}{#1}%
274 \GTS@j{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
275 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
276 }
277
278 \newcommand{\GTSxinc}[1][\GTSxi[#1]]{
279
280 \newcommand{\GTSxincidence}[1][\GTSxi[#1]]{

\GTSincidence This macro will draw an incidence symbol. The operand on the left of the symbol should be
\GTSinc considered the vertex to which the edge/the operand on the right of the symbol is incident.
\GTSi 281 \GTS@newcommand{\GTSincidence}{\GTS@i{}{}{}{}}
282 \GTS@newcommand{\GTSinc}{\GTS@i{}{}{}{}}
283 \GTS@newcommand{\GTSi}{\GTS@i{}{}{}{}}

\GTSincidenceRelation This macro will draw an incidence symbol with relation spacing in math mode.
\GTSincRel 284 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSi/}}
\GTSiR 285 \GTS@newcommand{\GTSincRel}{\mathrel{\GTSi/}}
286 \GTS@newcommand{\GTSiR}{\mathrel{\GTSi/}}

\GTSnegatedIncidence This macro will draw a negated incidence symbol.
\GTSnegInc 287 \GTS@newcommand{\GTSnegatedIncidence}{\GTS@i{}{}{1}{}{}}
\GTSnI 288 \GTS@newcommand{\GTSnegInc}{\GTS@i{}{}{1}{}{}}
289 \GTS@newcommand{\GTSnI}{\GTS@i{}{}{1}{}{}}

\GTSnegatedIncidence- This macro will draw a negated incidence symbol with relation spacing in math mode.
Relation 290 \GTS@newcommand{\GTSnegatedIncidenceRelation}{\mathrel{\GTSnI/}}
\GTSnegIncRel 291 \GTS@newcommand{\GTSnegIncRel}{\mathrel{\GTSnI/}}
\GTSnIR 292 \GTS@newcommand{\GTSnIR}{\mathrel{\GTSnI/}}

\GTSoutIncidence This macro will draw a directed incidence upward (out(ward) incidence) symbol. The
\GTSoutInc operand on the left of the symbol should be considered the vertex to which the arc/the
\GTSoI operand on the right of the symbol is incident.
293 \GTS@newcommand{\GTSoutIncidence}{\GTS@i{}{}{}{1}{}{}}
294 \GTS@newcommand{\GTSoutInc}{\GTS@i{}{}{}{1}{}{}}
295 \GTS@newcommand{\GTSoI}{\GTS@i{}{}{}{1}{}{}}

\GTSoutIncidenceRelation This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation
\GTSoutIncRel spacing in math mode.
\GTSoIR 296 \GTS@newcommand{\GTSoutIncidenceRelation}{\mathrel{\GTSoI/}}
297 \GTS@newcommand{\GTSoutIncRel}{\mathrel{\GTSoI/}}
298 \GTS@newcommand{\GTSoIR}{\mathrel{\GTSoI/}}

\GTSnegatedOutIncidence This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.
\GTSnegOutInc 299 \GTS@newcommand{\GTSnegatedOutIncidence}{\GTS@i{}{}{1}{1}{}{}}
\GTSnOI 300 \GTS@newcommand{\GTSnegOutInc}{\GTS@i{}{}{1}{1}{}{}}
301 \GTS@newcommand{\GTSnOI}{\GTS@i{}{}{1}{1}{}{}}

\GTSnegatedOutIncidence- This macro will draw a negated directed incidence upward (out(ward) incidence) symbol
Relation with relation spacing in math mode.
\GTSnegOutIncRel 302 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{\mathrel{\GTSnOI/}}
\GTSnOIR 303 \GTS@newcommand{\GTSnegOutIncRel}{\mathrel{\GTSnOI/}}
304 \GTS@newcommand{\GTSnOIR}{\mathrel{\GTSnOI/}}

\GTSinIncidence This macro will draw a directed incidence downward (in(ward) incidence) symbol. The
\GTSinInc operand on the left of the symbol should be considered the vertex to which the arc/the
\GTSiI operand on the right of the symbol is incident.
305 \GTS@newcommand{\GTSinIncidence}{\GTS@i{}{}{}{}{1}}
306 \GTS@newcommand{\GTSinInc}{\GTS@i{}{}{}{}{1}}
307 \GTS@newcommand{\GTSiI}{\GTS@i{}{}{}{}{1}}

```

`\GTSinIncidenceRelation` This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`\GTSinIncRel`

```

\GTSiIR 308 \GTS@newcommand{\GTSinIncidenceRelation}{\mathrel{\GTSiI/}}
309 \GTS@newcommand{\GTSinIncRel}{\mathrel{\GTSiI/}}
310 \GTS@newcommand{\GTSiIR}{\mathrel{\GTSiI/}}

```

`\GTSnegatedInIncidence` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

`\GTSnegInInc`

```

\GTSnII 311 \GTS@newcommand{\GTSnegatedInIncidence}{\GTS@i{}{}{1}{}{1}}
312 \GTS@newcommand{\GTSnegInInc}{\GTS@i{}{}{1}{}{1}}
313 \GTS@newcommand{\GTSnII}{\GTS@i{}{}{1}{}{1}}

```

`\GTSnegatedInIncidence-Relation` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`\GTSnegInIncRel`

```

\GTSnIIR 314 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{\mathrel{\GTSnII/}}
315 \GTS@newcommand{\GTSnegInIncRel}{\mathrel{\GTSnII/}}
316 \GTS@newcommand{\GTSnIIR}{\mathrel{\GTSnII/}}

```

`\GTSadjacencyMonochrome` This macro will draw a monochrome adjacency symbol.

`\GTSadjMono`

```

\GTSadjMon 317 \GTS@newcommand{\GTSadjacencyMonochrome}{\GTS@a{black}{black}{}{}{}}
318 \GTS@newcommand{\GTSadjMono}{\GTS@a{black}{black}{}{}{}}
\GTSaM 319 \GTS@newcommand{\GTSadjMon}{\GTS@a{black}{black}{}{}{}}
320 \GTS@newcommand{\GTSaM}{\GTS@a{black}{black}{}{}{}}

```

`\GTSadjacencyMonochrome-Relation` This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

`\GTSadjMonoRel`

```

\GTSadjMonRel 321 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{\mathrel{\GTSaM/}}
322 \GTS@newcommand{\GTSadjMonoRel}{\mathrel{\GTSaM/}}
323 \GTS@newcommand{\GTSadjMonRel}{\mathrel{\GTSaM/}}
\GTSaMR 324 \GTS@newcommand{\GTSaMR}{\mathrel{\GTSaM/}}

```

`\GTSunadjacencyMonochrome` This macro will draw a monochrome unadjacency symbol.

`\GTSunadjMono`

```

\GTSunadjMon 325 \GTS@newcommand{\GTSunadjacencyMonochrome}{\GTS@a{black}{black}{1}{}{}}
326 \GTS@newcommand{\GTSunadjMono}{\GTS@a{black}{black}{1}{}{}}
\GTSuM 327 \GTS@newcommand{\GTSunadjMon}{\GTS@a{black}{black}{1}{}{}}
328 \GTS@newcommand{\GTSuM}{\GTS@a{black}{black}{1}{}{}}

```

`\GTSunadjacencyMonochrome-Relation` This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

`\GTSunadjMonoRel`

```

\GTSunadjMonRel 329 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{\mathrel{\GTSuM/}}
330 \GTS@newcommand{\GTSunadjMonoRel}{\mathrel{\GTSuM/}}
\GTSuMR 331 \GTS@newcommand{\GTSunadjMonRel}{\mathrel{\GTSuM/}}
332 \GTS@newcommand{\GTSuMR}{\mathrel{\GTSuM/}}

```

`\GTSdirectedAdjacencyUp-Monochrome` This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

`\GTSdirAdjUpMono`

```

\GTSdirAdjUpMon 333 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
\GTSdAUM 334 \GTS@a{black}{black}{}{1}{}%
335 }
336 \GTS@newcommand{\GTSdirAdjUpMono}{\GTS@a{black}{black}{}{1}{}{}}
337 \GTS@newcommand{\GTSdirAdjUpMon}{\GTS@a{black}{black}{}{1}{}{}}
338 \GTS@newcommand{\GTSdAUM}{\GTS@a{black}{black}{}{1}{}{}}

```

`\GTSdirectedAdjacencyUp-MonochromeRelation` This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

`\GTSdirAdjUpMonoRel`

```

\GTSdirAdjUpMonRel 339 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
\GTSdAUMR 340 \mathrel{\GTSdAUM/}%
341 }
342 \GTS@newcommand{\GTSdirAdjUpMonoRel}{\mathrel{\GTSdAUM/}}
343 \GTS@newcommand{\GTSdirAdjUpMonRel}{\mathrel{\GTSdAUM/}}
344 \GTS@newcommand{\GTSdAUMR}{\mathrel{\GTSdAUM/}}

```

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency upward symbol.

`AdjacencyUpMonochrome` 345 `\GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%`
`\GTSnegDirAdjUpMono` 346 `\GTS@a{black}{black}{1}{1}{1}{1}{%}`
`\GTSnegDirAdjUpMon` 347 `}`
`\GTSnDAUM` 348 `\GTS@newcommand{\GTSnegDirAdjUpMono}{\GTS@a{black}{black}{1}{1}{1}{1}{%}`
349 `\GTS@newcommand{\GTSnegDirAdjUpMon}{\GTS@a{black}{black}{1}{1}{1}{1}{%}`
350 `\GTS@newcommand{\GTSnDAUM}{\GTS@a{black}{black}{1}{1}{1}{1}{%}`

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency upward symbol with relation
`AdjacencyUpMonochrome-` spacing in math mode.
`Relation` 351 `\GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%`
`\GTSnegDirAdjUpMonRel` 352 `\mathrel{\GTSnDAUM/}%}`
`\GTSnegDirAdjUpMonoRel` 353 `}`
`\GTSnDAUMR` 354 `\GTS@newcommand{\GTSnegDirAdjUpMonoRel}{\mathrel{\GTSnDAUM/}}`
355 `\GTS@newcommand{\GTSnegDirAdjUpMonRel}{\mathrel{\GTSnDAUM/}}`
356 `\GTS@newcommand{\GTSnDAUMR}{\mathrel{\GTSnDAUM/}}`

`\GTSdirectedAdjacencyDown-` This macro will draw a monochrome directed adjacency downward symbol. The operand on
`Monochrome` the left of the symbol should be considered the target of the arc; the operand on the right
`\GTSdirAdjDownMono` of the symbol should be considered the source of the arc.
`\GTSdirAdjDownMon` 357 `\GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%`
`\GTSdADM` 358 `\GTS@a{black}{black}{1}{1}{1}{1}{%}`
359 `}`
360 `\GTS@newcommand{\GTSdirAdjDownMono}{\GTS@a{black}{black}{1}{1}{1}{1}{%}`
361 `\GTS@newcommand{\GTSdirAdjDownMon}{\GTS@a{black}{black}{1}{1}{1}{1}{%}`
362 `\GTS@newcommand{\GTSdADM}{\GTS@a{black}{black}{1}{1}{1}{1}{%}`

`\GTSdirectedAdjacencyDown-` This macro will draw a monochrome directed adjacency downward symbol with relation
`MonochromeRelation` spacing in math mode.
`\GTSdirAdjDownMonoRel` 363 `\GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%`
`\GTSdirAdjDownMonRel` 364 `\mathrel{\GTSdADM/}%}`
`\GTSdADMR` 365 `}`
366 `\GTS@newcommand{\GTSdirAdjDownMonoRel}{\mathrel{\GTSdADM/}}`
367 `\GTS@newcommand{\GTSdirAdjDownMonRel}{\mathrel{\GTSdADM/}}`
368 `\GTS@newcommand{\GTSdADMR}{\mathrel{\GTSdADM/}}`

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency downward symbol.

`AdjacencyDownMonochrome` 369 `\GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%`
`\GTSnegDirAdjDownMono` 370 `\GTS@a{black}{black}{1}{1}{1}{1}{%}`
`\GTSnegDirAdjDownMon` 371 `}`
`\GTSnDADM` 372 `\GTS@newcommand{\GTSnegDirAdjDownMono}{\GTS@a{black}{black}{1}{1}{1}{1}{%}`
373 `\GTS@newcommand{\GTSnegDirAdjDownMon}{\GTS@a{black}{black}{1}{1}{1}{1}{%}`
374 `\GTS@newcommand{\GTSnDADM}{\GTS@a{black}{black}{1}{1}{1}{1}{%}`

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency downward symbol with
`AdjacencyDownMonochrome-` relation spacing in math mode.
`Relation` 375 `\GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%`
`\GTSnegDirAdjDownMonoRel` 376 `\mathrel{\GTSnDADM/}%}`
`\GTSnegDirAdjDownMonRel` 377 `}`
`\GTSnDADMR` 378 `\GTS@newcommand{\GTSnegDirAdjDownMonoRel}{\mathrel{\GTSnDADM/}}`
379 `\GTS@newcommand{\GTSnegDirAdjDownMonRel}{\mathrel{\GTSnDADM/}}`
380 `\GTS@newcommand{\GTSnDADMR}{\mathrel{\GTSnDADM/}}`

`\GTSincidenceMonochrome` This macro will draw a monochrome incidence symbol. The operand on the left of the
`\GTSincMono` symbol should be considered the vertex to which the edge/the operand on the right of the
`\GTSincMon` symbol is incident.
`\GTSiM` 381 `\GTS@newcommand{\GTSincidenceMonochrome}{\GTS@i{black}{black}{1}{1}{1}{1}{%}`
382 `\GTS@newcommand{\GTSincMono}{\GTS@i{black}{black}{1}{1}{1}{1}{%}`
383 `\GTS@newcommand{\GTSincMon}{\GTS@i{black}{black}{1}{1}{1}{1}{%}`
384 `\GTS@newcommand{\GTSiM}{\GTS@i{black}{black}{1}{1}{1}{1}{%}`

`\GTSincidenceMonochrome-` This macro will draw a monochrome incidence symbol with relation spacing in math mode.
`Relation` 385 `\GTS@newcommand{\GTSincidenceMonochromeRelation}{\mathrel{\GTSiM/}}`
`\GTSincMonoRel` 386 `\GTS@newcommand{\GTSincMonoRel}{\mathrel{\GTSiM/}}`
`\GTSincMonRel` 387 `\GTS@newcommand{\GTSincMonRel}{\mathrel{\GTSiM/}}`
`\GTSiMR` 388 `\GTS@newcommand{\GTSiMR}{\mathrel{\GTSiM/}}`

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol.

```

Monochrome 389 \GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%
\GTSnegIncMono 390 \GTS@i{black}{black}{1}{-}{-}%
\GTSnegIncMon 391 }
\GTSnIM 392 \GTS@newcommand{\GTSnegIncMono}{\GTS@i{black}{black}{1}{-}{-}}
393 \GTS@newcommand{\GTSnegIncMon}{\GTS@i{black}{black}{1}{-}{-}}
394 \GTS@newcommand{\GTSnIM}{\GTS@i{black}{black}{1}{-}{-}}

```

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

```

MonochromeRelation 395 \GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{\mathrel{\GTSnIM/}}
\GTSnegIncMonoRel 396 \GTS@newcommand{\GTSnegIncMonoRel}{\mathrel{\GTSnIM/}}
\GTSnIMR 397 \GTS@newcommand{\GTSnegIncMonRel}{\mathrel{\GTSnIM/}}
398 \GTS@newcommand{\GTSnIMR}{\mathrel{\GTSnIM/}}

```

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

\GTSoutIncMono 399 \GTS@newcommand{\GTSoutIncidenceMonochrome}{\GTS@i{black}{black}{-}{1}{-}}
400 \GTS@newcommand{\GTSoutIncMono}{\GTS@i{black}{black}{-}{1}{-}}
401 \GTS@newcommand{\GTSoutIncMon}{\GTS@i{black}{black}{-}{1}{-}}
402 \GTS@newcommand{\GTSoutIM}{\GTS@i{black}{black}{-}{1}{-}}

```

`\GTSoutIncidenceMonochrome-` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

Relation 403 \GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{\mathrel{\GTSoutIM/}}
\GTSoutIncMonoRel 404 \GTS@newcommand{\GTSoutIncMonoRel}{\mathrel{\GTSoutIM/}}
\GTSoutIncMonRel 405 \GTS@newcommand{\GTSoutIncMonRel}{\mathrel{\GTSoutIM/}}
406 \GTS@newcommand{\GTSoutIMR}{\mathrel{\GTSoutIM/}}

```

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

```

Monochrome 407 \GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%
\GTSnegOutIncMono 408 \GTS@i{black}{black}{1}{1}{-}%
\GTSnOIM 409 }
410 \GTS@newcommand{\GTSnegOutIncMono}{\GTS@i{black}{black}{1}{1}{-}}
411 \GTS@newcommand{\GTSnegOutIncMon}{\GTS@i{black}{black}{1}{1}{-}}
412 \GTS@newcommand{\GTSnOIM}{\GTS@i{black}{black}{1}{1}{-}}

```

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

MonochromeRelation 413 \GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%
\GTSnegOutIncMonoRel 414 \mathrel{\GTSnOIM/}%
\GTSnOIMR 415 }
416 \GTS@newcommand{\GTSnegOutIncMonoRel}{\mathrel{\GTSnOIM/}}
417 \GTS@newcommand{\GTSnegOutIncMonRel}{\mathrel{\GTSnOIM/}}
418 \GTS@newcommand{\GTSnOIMR}{\mathrel{\GTSnOIM/}}

```

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

\GTSinIncMono 419 \GTS@newcommand{\GTSinIncidenceMonochrome}{\GTS@i{black}{black}{-}{-}{1}}
420 \GTS@newcommand{\GTSinIncMono}{\GTS@i{black}{black}{-}{-}{1}}
421 \GTS@newcommand{\GTSinIncMon}{\GTS@i{black}{black}{-}{-}{1}}
422 \GTS@newcommand{\GTSiIM}{\GTS@i{black}{black}{-}{-}{1}}

```

`\GTSinIncidenceMonochrome-` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

Relation 423 \GTS@newcommand{\GTSinIncidenceMonochromeRelation}{\mathrel{\GTSiIM/}}
\GTSinIncMonoRel 424 \GTS@newcommand{\GTSinIncMonoRel}{\mathrel{\GTSiIM/}}
\GTSiIMR 425 \GTS@newcommand{\GTSinIncMonRel}{\mathrel{\GTSiIM/}}
426 \GTS@newcommand{\GTSiIMR}{\mathrel{\GTSiIM/}}

```



```

\GTSnegatedInIncidence- This macro will draw a monochrome negated directed incidence downward (in(ward) inci-
Monochrome dence) symbol.
\GTSnegInIncMono 427 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
\GTSnegInIncMon 428 \GTS@i{black}{black}{1}{1}%
\GTSnIIM 429 }
430 \GTS@newcommand{\GTSnegInIncMono}{\GTS@i{black}{black}{1}{1}}
431 \GTS@newcommand{\GTSnegInIncMon}{\GTS@i{black}{black}{1}{1}}
432 \GTS@newcommand{\GTSnIIM}{\GTS@i{black}{black}{1}{1}}

\GTSnegatedInIncidence- This macro will draw a monochrome negated directed incidence downward (in(ward) inci-
MonochromeRelation dence) symbol with relation spacing in math mode.
\GTSnegInIncMonoRel 433 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
\GTSnegInIncMonRel 434 \mathrel{\GTSnIIM/}%
\GTSnIIMR 435 }
436 \GTS@newcommand{\GTSnegInIncMonoRel}{\mathrel{\GTSnIIM/}}
437 \GTS@newcommand{\GTSnegInIncMonRel}{\mathrel{\GTSnIIM/}}
438 \GTS@newcommand{\GTSnIIMR}{\mathrel{\GTSnIIM/}}

439 \endinput
440 </package>

```

4 Acknowledgments

Thank You God! Thank You Father! Thank You Jesus! Thank You Holy-Spirit!

5 Support

This package is gratis and free. But if you can star it on GitHub, it is a nice encouragement.

6 Change History

v0.1.0 – 2026-08-01	v1.2.0 – 2026-09-06
General: Initial version 1	General: CTAN URL, new macros for same set of symbols 1
v1.0.0 – 2026-08-26	
General: First public release 1	v1.2.1 – 2026-09-08
v1.1.0 – 2026-08-29	General: smaller tests output 1
General: Added a memo 1	v1.3.0 – 2026-09-08
v1.1.1 – 2026-08-30	General: new macros for monochrome symbols 1
General: Small corrections and improvements 1	

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